

DDA3020 Machine Learning

Tutorial: Logistic Regression

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Instructions

This tutorial covers the core concepts of Logistic Regression, including model representation, sigmoid function, decision boundary, cross-entropy loss function, convexity analysis, regularized logistic regression, probabilistic perspective (Bernoulli distribution, MLE & MAP), the difference between linear regression and logistic regression, and multi-class classification extension (one-vs-all, Softmax regression). Show your working clearly for all calculation and derivation questions.

Useful Formulas

For a binary classification dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^m$ with $\mathbf{x}^{(i)} \in \mathbb{R}^d$, $y^{(i)} \in \{0, 1\}$:

- Sigmoid (Logistic) Function: $g(z) = \frac{1}{1+\exp(-z)}$, with derivative $g'(z) = g(z)(1-g(z))$
- Logistic Regression Hypothesis: $f_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}^\top \mathbf{x} + b) = P(y = 1|\mathbf{x}; \mathbf{w}) \in [0, 1]$ (bias b can be absorbed into $\mathbf{w}$ by augmenting $\mathbf{x}$ with a constant 1)
- Binary Cross-Entropy Loss: $J(\mathbf{w}) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$
- L2-Regularized Cross-Entropy Loss: $\bar{J}(\mathbf{w}) = J(\mathbf{w}) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2$ (bias term w_0 is not regularized)
- Gradient of Cross-Entropy Loss: $\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i) \mathbf{x}_i$
- Gradient Descent Update Rule: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$, where α is the learning rate

- Bernoulli Distribution PMF: $P(y|\mathbf{x}; \mathbf{w}) = f_{\mathbf{w}}(\mathbf{x})^y \cdot (1 - f_{\mathbf{w}}(\mathbf{x}))^{1-y}$
- Log-Likelihood Function: $\log \mathcal{L}(\mathbf{w}) = \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$
- Softmax Function (Multi-class Classification, C classes): $f_{\mathbf{w}_j}(\mathbf{x}) = \frac{\exp(\mathbf{w}_j^\top \mathbf{x})}{\sum_{c=1}^C \exp(\mathbf{w}_c^\top \mathbf{x})} = P(y = j|\mathbf{x}; W)$
- Softmax Regression Cross-Entropy Loss: $J(W) = -\frac{1}{m} \sum_{i=1}^m \sum_{j=1}^C \mathbb{I}(y_i = j) \log(f_{\mathbf{w}_j}(\mathbf{x}_i))$, where $\mathbb{I}(\cdot)$ is the indicator function

Exercises

Multiple Choice Questions

Q1. Which of the following is the core hypothesis output of binary logistic regression?

- (A) A continuous real value in $(-\infty, +\infty)$
- (B) The posterior probability $P(y = 1|\mathbf{x}; \mathbf{w}) \in [0, 1]$
- (C) A discrete class label without probability interpretation
- (D) The residual between prediction and ground truth

Q2. Which of the following is the correct expression of the sigmoid (logistic) function?

- (A) $g(z) = \frac{1}{1+\exp(z)}$
- (B) $g(z) = \frac{\exp(z)}{1+\exp(-z)}$
- (C) $g(z) = \frac{1}{1+\exp(-z)}$
- (D) $g(z) = \max(0, z)$

Q3. The standard cost function (loss) for binary logistic regression is:

- (A) Residual sum of squares (MSE loss): $\frac{1}{2m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2$
- (B) Cross-entropy loss: $-\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$
- (C) Hinge loss: $\max(0, 1 - y_i f_{\mathbf{w}}(\mathbf{x}_i))$
- (D) Mean absolute error (MAE loss): $\frac{1}{m} \sum_{i=1}^m |f_{\mathbf{w}}(\mathbf{x}_i) - y_i|$

Q4. Why is mean squared error (MSE) NOT suitable as the loss function for logistic regression?

- (A) MSE is not differentiable for logistic regression
- (B) MSE leads to a non-convex optimization problem with many local minima
- (C) MSE cannot penalize misclassified samples
- (D) MSE can only be used for regression tasks, not classification tasks

Q5. The decision boundary of binary logistic regression is determined by which of the following equations?

- (A) $\mathbf{w}^\top \mathbf{x} + b = 1$
- (B) $\mathbf{w}^\top \mathbf{x} + b = 0$

- (C) $f_{\mathbf{w}}(\mathbf{x}) = 0$
 - (D) $f_{\mathbf{w}}(\mathbf{x}) = 1$
- Q6. In L2-regularized logistic regression, which of the following parameters is NOT included in the regularization penalty term?
- (A) Feature weight w_1
 - (B) Feature weight w_d
 - (C) Bias term w_0 (intercept)
 - (D) All feature weights are unregularized
- Q7. From the probabilistic perspective, logistic regression assumes that the label y given input $\mathbf{x}$ follows which distribution?
- (A) Gaussian (Normal) distribution
 - (B) Bernoulli distribution
 - (C) Laplace distribution
 - (D) Uniform distribution
- Q8. Which of the following statements about the comparison between linear regression and logistic regression is INCORRECT?
- (A) Linear regression is for regression tasks, while logistic regression is for classification tasks
 - (B) Linear regression has a closed-form solution, while logistic regression typically uses gradient descent for optimization
 - (C) The hypothesis output of linear regression is unbounded, while that of logistic regression is bounded in $[0, 1]$
 - (D) Both linear regression and logistic regression use cross-entropy as the loss function
- Q9. In L2-regularized logistic regression, what is the impact of increasing the regularization parameter λ ?
- (A) The heavier the penalty on large weights, the easier the model is to overfit
 - (B) The heavier the penalty on large weights, the higher the model bias and the lower the variance
 - (C) The lighter the penalty on large weights, the easier the model is to underfit

- (D) λ controls the width of the decision boundary, with no impact on bias and variance
- Q10. For a multi-class classification task with C distinct classes, how many weight vectors need to be learned in standard Softmax regression (multi-class logistic regression)?
- (A) 1
- (B) $C - 1$
- (C) C
- (D) 2^C

Calculation Questions

- Q1. Calculate the output of the sigmoid function for $z = 0$, $z = 2$, and $z = -3$ respectively, retaining 3 decimal places.
- Q2. Given a sample $\mathbf{x} = (1, 3)$ with weight vector $\mathbf{w} = (0.5, -0.5)$ and bias $b = 0.5$, calculate the predicted probability $P(y = 1|\mathbf{x}; \mathbf{w})$ of logistic regression, retaining 3 decimal places.
- Q3. For a sample with true label $y = 1$, the logistic regression model gives a predicted probability $f_{\mathbf{w}}(\mathbf{x}) = 0.2$. Calculate the cross-entropy loss of this single sample, retaining 3 decimal places.
- Q4. Given a sample $\mathbf{x} = (2, 1)$, true label $y = 0$, weight vector $\mathbf{w} = (1, -1)$, bias $b = 0$. Calculate:
- (1) The cross-entropy loss of this sample;
 - (2) The gradient of the loss with respect to w_1 (the first dimension of $\mathbf{w}$).
- Q5. For L2-regularized logistic regression with $m = 5$ training samples, regularization parameter $\lambda = 4$, and weight vector $\mathbf{w} = (w_0 = 0, w_1 = 1, w_2 = 2, w_3 = 3)$. Calculate the value of the L2 regularization term in the objective function.

Short Answer Questions

- Q1. Briefly describe the role of the sigmoid function in logistic regression, and explain why it is suitable for binary classification tasks.

- Q2. Why does logistic regression use cross-entropy loss instead of mean squared error (MSE) loss? Explain from the perspective of optimization.
- Q3. Briefly describe the purpose of introducing regularization in logistic regression, and compare the different effects of L1 regularization and L2 regularization on model parameters.
- Q4. From the probabilistic perspective, explain the equivalence between the maximum likelihood estimation (MLE) of logistic regression and the minimization of cross-entropy loss.
- Q5. Compare linear regression and logistic regression from four dimensions: task type, model hypothesis, loss function, and solution method.
- Q6. What is the decision boundary of logistic regression? How to obtain a nonlinear decision boundary with logistic regression?
- Q7. Briefly describe the core idea of the one-vs-all (one-vs-rest) strategy for multi-class classification with logistic regression, and its difference from Softmax regression.

Derivation Questions

- Q1. For binary logistic regression, derive the cross-entropy loss function from the maximum likelihood estimation (MLE) of the Bernoulli distribution, and write the complete derivation steps.
- Q2. Derive the gradient of the binary logistic regression cross-entropy loss function with respect to the weight vector $\mathbf{w}$, and write the gradient descent update formula.
- Q3. Prove that L2-regularized logistic regression is equivalent to the maximum a posteriori (MAP) estimation with a Gaussian prior on the weight vector $\mathbf{w}$, and write the complete derivation steps.

Answers & Detailed Solutions

Multiple Choice Questions

Q1. **Answer: B**

Explanation: Binary logistic regression models the posterior probability $P(y = 1 \mid \mathbf{x}; \mathbf{w})$ using the sigmoid function, which outputs a value in $[0, 1]$ representing the probability of the positive class.

Q2. **Answer: C**

Explanation: The standard sigmoid (logistic) function is defined as $g(z) = \frac{1}{1 + \exp(-z)}$. It maps any real-valued z to the interval $(0, 1)$.

Q3. **Answer: B**

Explanation: The cross-entropy loss (also called log loss) is derived from the principle of maximum likelihood estimation and is convex for logistic regression, making it the standard choice.

Q4. **Answer: B**

Explanation: Applying mean squared error to the sigmoid output produces a non-convex loss surface with many local minima, which complicates gradient-based optimization. Cross-entropy avoids this issue.

Q5. **Answer: B**

Explanation: Logistic regression predicts $y = 1$ when $f_{\mathbf{w}}(\mathbf{x}) \geq 0.5$, which is equivalent to $\mathbf{w}^\top \mathbf{x} + b \geq 0$. The decision boundary is the threshold where $f_{\mathbf{w}}(\mathbf{x}) = 0.5$, i.e., $\mathbf{w}^\top \mathbf{x} + b = 0$.

Q6. **Answer: C**

Explanation: In L2-regularized logistic regression, the bias (intercept) term w_0 is typically excluded from regularization. Penalizing the intercept would force the decision boundary unnecessarily through the origin.

Q7. **Answer: B**

Explanation: For a binary classification task, the label $y \in \{0, 1\}$ is modeled as a Bernoulli random variable whose success probability $p = P(y = 1 \mid \mathbf{x})$ is given by the logistic function.

Q8. **Answer: D**

Explanation: Linear regression uses mean squared error (or similar residual-based losses), not cross-entropy. Cross-entropy is specifically designed for classification tasks where the output represents a probability.

Q9. **Answer: B**

Explanation: Increasing the regularization parameter λ imposes a stronger penalty on large feature weights, reducing model complexity. This typically increases bias but decreases variance, helping to prevent overfitting.

Q10. **Answer: C**

Explanation: In Softmax regression (multinomial logistic regression), a separate weight vector is learned for each of the C classes. The output of each vector is passed through the softmax function to produce class probabilities.

Calculation Questions

Q1. **Answer:** $g(0) = 0.500$, $g(2) \approx 0.881$, $g(-3) \approx 0.047$

Explanation:

$$\begin{aligned}g(0) &= \frac{1}{1 + \exp(0)} = \frac{1}{2} = 0.5 \\g(2) &= \frac{1}{1 + \exp(-2)} \approx \frac{1}{1 + 0.1353} \approx 0.881 \\g(-3) &= \frac{1}{1 + \exp(3)} \approx \frac{1}{1 + 20.0855} \approx 0.047\end{aligned}$$

Q2. **Answer:** ≈ 0.378

Explanation: First calculate the linear output: $z = \mathbf{w}^\top \mathbf{x} + b = 0.5 \times 1 + (-0.5) \times 3 + 0.5 = 0.5 - 1.5 + 0.5 = -0.5$

Then apply sigmoid: $g(-0.5) = \frac{1}{1 + \exp(0.5)} \approx \frac{1}{1 + 1.6487} \approx 0.378$

Q3. **Answer:** ≈ 1.609

Explanation: For $y = 1$, cross-entropy loss is $-\log(f_{\mathbf{w}}(\mathbf{x})) = -\log(0.2) \approx 1.609$

Q4. **Answer:** (1) ≈ 1.313 ; (2) ≈ 1.462

Explanation:

(a) Linear output $z = 1 \times 2 + (-1) \times 1 + 0 = 1$, predicted probability $f_{\mathbf{w}}(\mathbf{x}) = g(1) = \frac{1}{1 + e^{-1}} \approx 0.73106$. For $y = 0$, loss is $-\log(1 - f_{\mathbf{w}}(\mathbf{x})) = -\log(1 - 0.73106) = -\log(0.26894) \approx 1.313$.

- (b) Gradient of cross-entropy loss with respect to w_j is $(f_{\mathbf{w}}(\mathbf{x}) - y)x_j$. Here $f_{\mathbf{w}}(\mathbf{x}) - y = 0.73106 - 0 = 0.73106$, $x_1 = 2$, so gradient is $0.73106 \times 2 \approx 1.462$.

Q5. **Answer:** 5.6

Explanation: The L2 regularization term for logistic regression is $\frac{\lambda}{2m} \sum_{j=1}^d w_j^2$ (bias w_0 is excluded). Sum of squared weights: $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$. Regularization term: $\frac{4}{2 \times 5} \times 14 = \frac{4}{10} \times 14 = 0.4 \times 14 = 5.6$.

Short Answer Questions

Q1. **Answer:**

The core role of the sigmoid function in logistic regression is to map the unbounded linear output $z = \mathbf{w}^\top \mathbf{x}$ (range $(-\infty, +\infty)$) to a bounded value in $[0, 1]$, which is interpreted as the posterior probability $P(y = 1 | \mathbf{x}; \mathbf{w})$ of the sample belonging to the positive class.

It is suitable for binary classification for the following reasons emphasized in the lecture:

- (a) **Valid probability range:** Its output is strictly bounded in $[0, 1]$, which conforms to the definition of probability for binary classification.
- (b) **Decision boundary compatibility:** It naturally maps to the binary decision threshold: $f_{\mathbf{w}}(\mathbf{x}) \geq 0.5$ (predict $y = 1$) is equivalent to $\mathbf{w}^\top \mathbf{x} \geq 0$, which defines a clear decision boundary.
- (c) **Compatibility with probabilistic modeling:** It aligns with the Bernoulli distribution assumption of logistic regression, enabling maximum likelihood estimation for model training.

Additional Benefits:

- (a) It is a strictly monotonically increasing and continuously differentiable function, which is convenient for gradient-based optimization algorithms.
- (b) It has a smooth S-shaped curve with a large gradient near the threshold, making the model sensitive to changes near the classification boundary.

Q2. **Answer:**

The core reason emphasized in the lecture is as follows:

- (a) **Convexity guarantee:** The cross-entropy loss is a convex function with respect to the model parameters $\mathbf{w}$, which ensures a unique global optimal solution for the optimization problem. In contrast, the MSE loss combined with

the sigmoid function results in a non-convex objective function with multiple local minima, which cannot guarantee convergence to the global optimum.

Additional Benefits:

- (a) **Avoid gradient saturation:** For MSE loss, the gradient contains the derivative of the sigmoid function $g(z)(1 - g(z))$, which approaches 0 when $|z|$ is large, leading to gradient vanishing. For cross-entropy loss, the gradient is proportional to the prediction error $(f_{\mathbf{w}}(\mathbf{x}) - y)$, which will not vanish even for high-confidence samples, ensuring effective parameter updates.

Q3. Answer:

The core purpose of introducing regularization in logistic regression is to address the overfitting problem. Regularization keeps all features but reduces the magnitude of model parameters, thus reducing model complexity and improving the generalization ability on unseen data.

Key differences between L1 and L2 regularization emphasized in the lecture:

- (a) **Probabilistic prior:** L2 regularization corresponds to a Gaussian prior on the weight vector, while L1 regularization corresponds to a Laplace prior.
- (b) **Penalty form:** L2 regularization adds a penalty term proportional to the squared L2 norm of weights $\lambda \|\mathbf{w}\|_2^2$, while L1 regularization adds a penalty proportional to the L1 norm of weights $\lambda \|\mathbf{w}\|_1$.
- (c) **Effect on parameters:** L2 regularization shrinks all parameter values towards 0, while L1 regularization tends to produce sparse parameter solutions (shrinking unimportant feature weights to exactly 0).
- (d) **Bias term handling:** Both L1 and L2 regularization do not penalize the bias term w_0 .

Additional Benefits:

- (a) L2 regularization is continuously differentiable everywhere, which is more convenient for gradient-based optimization, and the solution is unique and stable.
- (b) L1 regularization can achieve feature selection while preventing overfitting, which is useful for high-dimensional data with many irrelevant features.

Q4. Answer:

The equivalence is derived based on the probabilistic modeling of logistic regression emphasized in the lecture, with the following core steps:

- (a) **Distribution assumption:** Logistic regression assumes that the label y given input $\mathbf{x}$ follows a Bernoulli distribution: $y(\mathbf{x}; \mathbf{w}) \sim \text{Bernoulli}(\mu(\mathbf{x}; \mathbf{w}))$, where $\mu(\mathbf{x}; \mathbf{w}) = \text{Sigmoid}(\mathbf{w}^\top \mathbf{x})$.
- (b) **Probability mass function:** The conditional probability can be written as $P(y|\mathbf{x}; \mathbf{w}) = \mu^y \cdot (1 - \mu)^{1-y}$.
- (c) **Log-likelihood function:** For m independent samples, the log-likelihood of parameter $\mathbf{w}$ is:

$$\log \mathcal{L}(\mathbf{w}) = \sum_{i=1}^m [y_i \log \mu(\mathbf{x}_i; \mathbf{w}) + (1 - y_i) \log(1 - \mu(\mathbf{x}_i; \mathbf{w}))]$$

- (d) **Equivalence:** Maximum likelihood estimation (MLE) aims to maximize $\log \mathcal{L}(\mathbf{w})$, which is exactly equivalent to minimizing the negative log-likelihood. The cross-entropy loss function of logistic regression is the average negative log-likelihood:

$$J(\mathbf{w}) = -\frac{1}{m} \log \mathcal{L}(\mathbf{w})$$

Thus, maximizing the log-likelihood is equivalent to minimizing the cross-entropy loss.

Additional Benefits: The constant factor $\frac{1}{m}$ does not change the optimal solution of the optimization problem, as it only scales the loss function uniformly.

Q5. Answer:

The comparison between linear regression and logistic regression summarized in the lecture is as follows:

Additional Benefits: Both models can be derived from both deterministic and probabilistic perspectives, and both are special cases of generalized linear models.

Q6. Answer:

The decision boundary of logistic regression is the hypersurface that divides the feature space into positive and negative class regions. As emphasized in the lecture, the decision boundary corresponds to the set of points where $f_{\mathbf{w}}(\mathbf{x}) = 0.5$, which is equivalent to $\mathbf{w}^\top \mathbf{x} = 0$. For linear features, the decision boundary is a linear hyperplane.

To obtain a nonlinear decision boundary with logistic regression, we use basis expansion (feature transformation): map the original input features $\mathbf{x}$ to a high-dimensional nonlinear feature space via a nonlinear basis function $\phi(\mathbf{x})$ (e.g., polynomial terms), then train the logistic regression model on the transformed features

Dimension	Linear Regression	Logistic Regression
Task Type	Regression task, predicting continuous real values	Classification task, predicting discrete class labels
Model Hypothesis	Linear function: $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$ Output range: $(-\infty, +\infty)$	Sigmoid-mapped linear function: $f_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}^\top \mathbf{x})$ Output range: $[0, 1]$ (posterior probability of positive class)
Loss Function	Residual sum of squares (MSE loss): $\frac{1}{2m} \sum_{i=1}^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$	Cross-entropy loss: $-\frac{1}{m} \sum_{i=1}^m \left[y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i)) \right]$
Solution Method	Closed-form solution: $\mathbf{w}^* = (X^\top X)^{-1} X^\top \mathbf{y}$ or gradient descent	No closed-form solution, optimized via gradient descent

$\phi(\mathbf{x})$. The decision boundary $\mathbf{w}^\top \phi(\mathbf{x}) = 0$ in the transformed space corresponds to a nonlinear boundary in the original feature space.

Additional Benefits: This method retains the linearity of the logistic regression model in the transformed feature space, while enabling it to fit complex nonlinear classification patterns.

Q7. Answer:

Core idea of the one-vs-all (one-vs-rest) strategy emphasized in the lecture: For a multi-class classification task with C classes, train a binary logistic regression classifier $f_{\mathbf{w}_j}(\cdot)$ for each class j , by treating all samples of class j as the positive class and all samples of other classes as the negative class. For a new test sample, predict its class as $\arg \max_j f_{\mathbf{w}_j}(\mathbf{x})$. Its pros are easy to implement, while cons are high training cost and difficulty in scaling to tasks with a large number of classes.

Key differences from Softmax regression emphasized in the lecture:

- (a) **Model structure:** One-vs-all trains C independent binary logistic regression models, while Softmax regression is a single unified multi-class model that jointly learns C weight vectors.
- (b) **Output property:** Softmax regression outputs a normalized probability distribution via the Softmax function, where the sum of predicted probabilities

of all C classes is exactly 1, while the outputs of one-vs-all classifiers are independent probabilities with no sum constraint.

- (c) **Optimization:** Softmax regression optimizes a unified multi-class cross-entropy loss, with all weight vectors updated in parallel, while one-vs-all optimizes each binary classifier independently.

Additional Benefits: One-vs-all is suitable for scenarios where classes are not completely mutually exclusive, while Softmax regression is more suitable for standard multi-class tasks where classes are mutually exclusive.

Derivation Questions

Q1. **Answer:**

Step 1: Model and distribution assumption

For binary logistic regression, we assume that the label y_i of sample $\mathbf{x}_i$ follows a Bernoulli distribution, with the probability of $y_i = 1$ being:

$$\mu_i = f_{\mathbf{w}}(\mathbf{x}_i) = g(\mathbf{w}^\top \mathbf{x}_i + b) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{x}_i + b))}$$

The probability mass function of the Bernoulli distribution can be written in a unified form:

$$P(y_i | \mathbf{x}_i; \mathbf{w}, b) = \mu_i^{y_i} \cdot (1 - \mu_i)^{1-y_i}$$

This formula holds for both $y_i = 1$ and $y_i = 0$: when $y_i = 1$, it simplifies to μ_i ; when $y_i = 0$, it simplifies to $1 - \mu_i$.

Step 2: Construct the likelihood function

For a training set with m independent and identically distributed samples, the likelihood function of the parameters $\mathbf{w}$ and b is the product of the probability of each sample:

$$\mathcal{L}(\mathbf{w}, b) = \prod_{i=1}^m P(y_i | \mathbf{x}_i; \mathbf{w}, b) = \prod_{i=1}^m [\mu_i^{y_i} \cdot (1 - \mu_i)^{1-y_i}]$$

Step 3: Take the log-likelihood

Since the logarithm is a monotonically increasing function, maximizing the likelihood function is equivalent to maximizing the log-likelihood function. Taking the natural logarithm of both sides converts the product into a sum, which simplifies the calculation:

$$\log \mathcal{L}(\mathbf{w}, b) = \sum_{i=1}^m [y_i \log \mu_i + (1 - y_i) \log(1 - \mu_i)]$$

Step 4: Equivalent to minimizing cross-entropy loss

Maximum likelihood estimation aims to maximize $\log \mathcal{L}(\mathbf{w}, b)$, which is completely equivalent to minimizing the negative log-likelihood. We define the cross-entropy loss function as the average negative log-likelihood (divided by the sample size m , a constant factor that does not affect the optimal solution):

$$J(\mathbf{w}, b) = -\frac{1}{m} \log \mathcal{L}(\mathbf{w}, b)$$

Substitute the log-likelihood expression to get the standard cross-entropy loss function of logistic regression:

$$J(\mathbf{w}, b) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$$

Thus, the maximum likelihood estimation of logistic regression is completely equivalent to the minimization of cross-entropy loss.

Q2. Answer:

Step 1: Define the loss function and sigmoid derivative

For simplicity, we incorporate the bias b into the weight vector $\mathbf{w}$ (augment $\mathbf{x}_i$ with a constant 1), so the model output is $f_{\mathbf{w}}(\mathbf{x}_i) = g(z_i)$, where $z_i = \mathbf{w}^\top \mathbf{x}_i$, and $g(z) = \frac{1}{1 + \exp(-z)}$ is the sigmoid function.

First, derive the derivative of the sigmoid function:

$$g'(z) = \frac{d}{dz} \left(\frac{1}{1 + \exp(-z)} \right) = \frac{\exp(-z)}{(1 + \exp(-z))^2} = g(z) \cdot (1 - g(z))$$

This is a key property of the sigmoid function, which simplifies the gradient derivation.

The cross-entropy loss function for a single sample is:

$$\ell_i(\mathbf{w}) = -[y_i \log(g(z_i)) + (1 - y_i) \log(1 - g(z_i))]$$

The overall loss is $J(\mathbf{w}) = \frac{1}{m} \sum_{i=1}^m \ell_i(\mathbf{w})$.

Step 2: Derive the gradient of the single-sample loss with respect to $\mathbf{w}$

Using the chain rule:

$$\frac{\partial \ell_i}{\partial \mathbf{w}} = \frac{\partial \ell_i}{\partial g(z_i)} \cdot \frac{\partial g(z_i)}{\partial z_i} \cdot \frac{\partial z_i}{\partial \mathbf{w}}$$

Calculate each term separately:

$$\begin{aligned} \frac{\partial \ell_i}{\partial g(z_i)} &= - \left[\frac{y_i}{g(z_i)} - \frac{1 - y_i}{1 - g(z_i)} \right] \\ \frac{\partial g(z_i)}{\partial z_i} &= g(z_i)(1 - g(z_i)) \\ \frac{\partial z_i}{\partial \mathbf{w}} &= \mathbf{x}_i \end{aligned}$$

Multiply the three terms together:

$$\begin{aligned}
\frac{\partial \ell_i}{\partial \mathbf{w}} &= - \left[\frac{y_i}{g(z_i)} - \frac{1 - y_i}{1 - g(z_i)} \right] \cdot g(z_i)(1 - g(z_i)) \cdot \mathbf{x}_i \\
&= - [y_i(1 - g(z_i)) - (1 - y_i)g(z_i)] \cdot \mathbf{x}_i \\
&= - [y_i - y_i g(z_i) - g(z_i) + y_i g(z_i)] \cdot \mathbf{x}_i \\
&= - [y_i - g(z_i)] \cdot \mathbf{x}_i \\
&= (g(z_i) - y_i) \cdot \mathbf{x}_i
\end{aligned}$$

Step 3: Derive the gradient of the overall loss

The gradient of the average loss $J(\mathbf{w})$ with respect to $\mathbf{w}$ is the average of the single-sample gradients:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = \frac{1}{m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i) \mathbf{x}_i$$

Step 4: Gradient descent update formula

In gradient descent, we update the weight vector along the negative direction of the gradient, with learning rate α :

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \cdot \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}}$$

Substitute the gradient expression to get the final update formula:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i) \mathbf{x}_i$$

Q3. **Answer:**

Step 1: Bayesian estimation framework

In Bayesian estimation, the maximum a posteriori (MAP) estimation of the parameter $\mathbf{w}$ is obtained by maximizing the posterior probability $P(\mathbf{w}|D)$, where D is the training dataset. According to Bayes' theorem:

$$P(\mathbf{w}|D) = \frac{P(D|\mathbf{w}) \cdot P(\mathbf{w})}{P(D)}$$

Since $P(D)$ is a constant independent of $\mathbf{w}$, maximizing the posterior probability is equivalent to maximizing the numerator $P(D|\mathbf{w}) \cdot P(\mathbf{w})$, i.e.:

$$\mathbf{w}_{MAP} = \arg \max_{\mathbf{w}} [\log P(D|\mathbf{w}) + \log P(\mathbf{w})]$$

Where $P(D|\mathbf{w})$ is the likelihood function of the data, and $P(\mathbf{w})$ is the prior distribution of the parameter $\mathbf{w}$.

Step 2: Likelihood function and prior distribution

- (a) **Likelihood function:** As derived in Q1, for logistic regression, the log-likelihood of the data is:

$$\log P(D|\mathbf{w}) = \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$$

- (b) **Gaussian prior assumption:** We assume that the weight vector $\mathbf{w}$ (excluding the bias term) follows a zero-mean Gaussian prior distribution, with each dimension independent and identically distributed:

$$w_j \sim \mathcal{N}(0, \tau^2), \quad j = 1, 2, \dots, d$$

The probability density function of the prior is:

$$P(\mathbf{w}) = \prod_{j=1}^d \frac{1}{\sqrt{2\pi\tau}} \exp\left(-\frac{w_j^2}{2\tau^2}\right)$$

Take the natural logarithm of the prior:

$$\log P(\mathbf{w}) = \sum_{j=1}^d \left[\log\left(\frac{1}{\sqrt{2\pi\tau}}\right) - \frac{w_j^2}{2\tau^2} \right] = -\frac{d}{2} \log(2\pi\tau^2) - \frac{1}{2\tau^2} \sum_{j=1}^d w_j^2$$

The first term $-\frac{d}{2} \log(2\pi\tau^2)$ is a constant independent of $\mathbf{w}$, which does not affect the optimization result.

Step 3: MAP optimization objective

Substitute the log-likelihood and log-prior into the MAP optimization objective:

$$\mathbf{w}_{MAP} = \arg \max_{\mathbf{w}} \left[\sum_{i=1}^m (y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))) - \frac{1}{2\tau^2} \sum_{j=1}^d w_j^2 + C \right]$$

Where C is the constant term from the prior. Maximizing the above objective is equivalent to minimizing the negative of the objective (removing the constant term):

$$\mathbf{w}_{MAP} = \arg \min_{\mathbf{w}} \left[-\sum_{i=1}^m (y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))) + \frac{1}{2\tau^2} \sum_{j=1}^d w_j^2 \right]$$

Step 4: Equivalence with L2-regularized logistic regression

Divide the entire objective function by the sample size m (a constant factor that does not change the optimal solution), and define the regularization parameter $\lambda = \frac{m}{\tau^2}$:

$$\mathbf{w}_{MAP} = \arg \min_{\mathbf{w}} \left[-\frac{1}{m} \sum_{i=1}^m (y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2 \right]$$

This is exactly the objective function of L2-regularized logistic regression. Therefore, L2-regularized logistic regression is equivalent to the MAP estimation of the weight vector with a zero-mean Gaussian prior.